Debt & Equity Instruments

Bonds & Stocks

Meaning, Features & Terminologies

Coupon Rate • Nominal Value • Redemption • Yield to Maturity • Dividend Growth

All figures illustrative, in Indian Rupees (₹), for teaching purposes.

FOUNDATIONS

Features of Debt as a Source of Finance

1

Contractual Obligation

Issuer must pay interest & repay principal regardless of profit.

2

Fixed Periodic Payments

Coupon is pre-set — unlike discretionary equity dividends.

3

Priority Over Equity

Debt holders are paid before shareholders, even on liquidation.

4

No Ownership / Voting

Bondholders are creditors, not owners of the company.

5

Finite Maturity (usually)

Most debt has a fixed repayment date (perpetual bonds excepted).

6

Tax-Deductible Interest

Interest lowers taxable income — the 'tax shield' on debt.

Real-world illustration: When Reliance Industries Ltd issued ₹20,000 crore of secured NCDs (face value ₹1,00,000 each) in Nov 2023, it took on a fixed obligation to pay interest and repay the principal by 2033 — with no ownership stake given to investors.

FOUNDATIONS

Meaning of a Bond — for an Indian Investor

A bond is a debt security issued by a company, bank, PSU, or the Government of India to raise long-term funds. The issuer promises to pay the bondholder a fixed periodic coupon (interest) and to repay the face value (principal) on a specified maturity / redemption date.

Think of it as an IOU — you lend ₹1,000 to an Indian issuer today; it promises fixed interest and to return your ₹1,000 at maturity.

Government of India

G-Secs, Treasury Bills

PSUs

NHAI · REC · PFC · IRFC

Banks

SBI · HDFC Bank · ICICI (AT1 / Tier-2)

Corporates

Reliance · Tata Steel · M&M (NCDs)

Key Bond Terminologies

Term	Meaning
Nominal / Face Value (F)	Value printed on the bond (₹100 / ₹1,000 / ₹1,00,000) — also the amount repaid at redemption.
Coupon Rate	Stated annual interest rate applied to face value, e.g. 8% coupon rate on ₹1,000 face value.
Coupon (C)	Actual rupee interest paid each period — Coupon = Coupon Rate × Face Value (e.g. ₹80).
Redemption / Maturity Value	Amount paid to the bondholder at maturity — usually equal to face value (at par).
Maturity Date	The date on which redemption occurs and the bond ceases to exist.
Market Price	Current traded price — may sit at a discount, premium, or par to face value.
Yield to Maturity (YTM)	Total return earned if bought at market price and held to redemption.

BONDS

Coupon (Plain Vanilla) Bond

$$P = C/(1+r)^1 + C/(1+r)^2 + \ldots + (C + F)/(1+r)^T$$

Present value of periodic coupons plus the face value repaid at maturity

A **plain vanilla bond** pays a fixed coupon each year and returns the full nominal (face) value on the redemption date.

% Coupon Rate vs YTM

If the coupon rate is below the market's required yield, the bond trades at a discount — and vice-versa.

₹ Redemption is Fixed

Regardless of the price paid, the investor always receives the full-face value at maturity.

Coupon Bond: Examples

EXAMPLE 1

Tata Steel Bond (Discount)

- Face value, F = ₹1,000
- Coupon rate 8% ⟶ C = ₹80 / yr
- Maturity, T = 3 years
- Required yield, r = 9%

$P = 80/1.09 + 80/1.09^2 + 1080/1.09^3$

$P = 73.39 + 67.34 + 833.83$

Price = ₹974.56

Coupon rate (8%) < YTM (9%) ⟶ bond trades at a discount; ₹1,000 is still repaid at redemption.

EXAMPLE 2

Mahindra & Mahindra Bond (Premium)

- Face value, F = ₹1,000
- Coupon rate 10% ⟶ C = ₹100 / yr
- Maturity, T = 2 years
- Required yield, r = 8%

$P = 100/1.08 + 1100/1.08^2$

$P = 92.59 + 943.07$

Price = ₹1,035.66

Coupon rate (10%) > YTM (8%) ⟶ bond trades at a premium; redemption value is still ₹1,000.

BONDS

Zero-Coupon Bond

$$P = F / (1+r)^T$$

Price is simply the present value of the redemption (face) value

A zero-coupon bond pays **no periodic interest**. It is bought at a deep discount, and the entire investor return comes from the gap between purchase price and the nominal value received at redemption.

1

No Coupon Rate

There is no periodic coupon at all — hence 'zero-coupon'.

2

No Reinvestment Risk

No coupons to reinvest, unlike a regular coupon bond.

3

Full Return at Redemption

Return = Redemption value − Purchase price.

Zero-Coupon Bond: Examples

EXAMPLE 1

NHAI Zero-Coupon Bond

- Nominal (redemption) value, F = ₹1,00,000
- Required yield, r = 7% p.a.
- Maturity, T = 5 years

$P = 1{,}00{,}000 / (1.07)^5$

$P = 1{,}00{,}000 / 1.4026$

Price ≈ ₹71,299

Investor pays ₹71,299 today; no coupons; receives full ₹1,00,000 at redemption after 5 years (gain ≈ ₹28,701).

EXAMPLE 2

REC Ltd Zero-Coupon Bond

- Nominal (redemption) value, F = ₹10,000
- Required yield, r = 8% p.a.
- Maturity, T = 10 years

$P = 10{,}000 / (1.08)^{10}$

$P = 10{,}000 / 2.1589$

Price ≈ ₹4,632

Useful for goal-based saving — buy at ₹4,632, receive ₹10,000 after 10 years, with nothing to reinvest in between.

BONDS

Perpetual Bond

$$P = C / r$$

Present value of a coupon stream that never stops

A perpetual bond has **no maturity date** — the issuer pays the coupon forever (subject to a call option) and the **face value is never redeemed**.

In India: Additional Tier-1 (AT1) Bonds

Issued by banks such as SBI, HDFC Bank, ICICI Bank, and PNB under Basel III norms. AT1 bonds rank just above equity and can even be written down under regulatory stress — as happened with Yes Bank's AT1 bonds in 2020 (~₹8,400 crore written off on RBI's direction). This shows that a bond's safety depends on where it sits in the capital structure — not just on being called a 'bond'.

Perpetual Bond: Examples

EXAMPLE 1

Bank of Baroda AT1 (Perpetual) Bond

- Face value, F = ₹100 (never redeemed)
- Coupon rate 8% ⟶ C = ₹8 / yr, forever
- Required yield, r = 8.5%

P = 8 / 0.085

Price ≈ ₹94.12

No maturity date, no redemption of the ₹100 face value — only the perpetual coupon stream is priced.

EXAMPLE 2

Power Finance Corp (PFC) Perpetual Bond

- Face value, F = ₹100 (never redeemed)
- Coupon rate 9% ⟶ C = ₹9 / yr, forever
- Required yield, r = 9.5%

P = 9 / 0.095

Price ≈ ₹94.74

'Redemption value' plays no role in the formula at all — only the coupon and required yield matter.

Yield to Maturity (YTM)

$$YTM \approx [\,C + (F - P)/n\,] \;/\; [\,(F + P)/2\,]$$

A quick hand-calculation approximation of the true (IRR-based) yield

YTM is the total return an investor earns if the bond is bought at its **current market price** and held to redemption — combining coupon income and any capital gain or loss versus the nominal value.

↑ Price below face value

YTM > coupon rate — investor gains on redemption.

↓ Price above face value

YTM < coupon rate — investor loses a little on redemption.

Yield to Maturity: Examples

EXAMPLE 1

Infosys-style Corporate Bond

- Face value, F = ₹1,000
- Coupon rate 9% ⟶ C = ₹90
- Years to maturity, n = 2
- Market price, P = ₹980

YTM ≈ [90 + (1000−980)/2] / [(1000+980)/2]

YTM ≈ 100 / 990

YTM ≈ 10.10%

Coupon rate is 9%, but true YTM is ~10.10% — bought below face value, so YTM > coupon rate.

EXAMPLE 2

L&T-style Corporate Bond

- Face value, F = ₹1,000
- Coupon rate 7% ⟶ C = ₹70
- Years to maturity, n = 3
- Market price, P = ₹950

YTM ≈ [70 + (1000−950)/3] / [(1000+950)/2]

YTM ≈ 86.67 / 975

YTM ≈ 8.89%

Buying at a ₹50 discount to nominal value lifts YTM well above the stated 7% coupon rate.



PART TWO

Stock (Equity) Valuation Concepts

Using the Dividend Discount Model — dividends play the role of coupons, but equity has no redemption value.

EQUITY

Zero Growth Dividend Discount Model

$$P_0 = D_1 / r$$

Dividend assumed constant forever — suited to mature, no-growth firms

EXAMPLE 1

Coal India Stock

- Expected annual dividend, D_1 = ₹18 (constant forever)
- Required return on equity, r = 12%

$P_0 = 18 / 0.12$

Price = ₹150 per share

Stable, high-payout PSU — nearly all profit distributed rather than reinvested for growth.

EXAMPLE 2

NTPC Stock

- Expected annual dividend, D_1 = ₹9 (constant forever)
- Required return on equity, r = 10%

$P_0 = 9 / 0.10$

Price = ₹90 per share

Model suits mature utilities with flat, dependable dividend payouts and minimal growth.

EQUITY

Constant Growth (Gordon Growth) Model

$$P_0 = D_1 / (r - g) \quad \text{where } D_1 = D_0(1+g)$$

Fits stable companies growing at a modest, sustainable rate forever

EXAMPLE 1

HDFC Bank Stock

- Last dividend, D_0 = ₹19
- Growth rate, g = 10%
- Required return, r = 14%

$D_1 = 19 \times 1.10 = ₹20.90$

$P_0 = 20.90 / (0.14 - 0.10)$

Price = ₹522.50 per share

Only valid while r > g — breaks down for high-growth-phase companies.

EXAMPLE 2

Asian Paints Stock

- Last dividend, D_0 = ₹15
- Growth rate, g = 8%
- Required return, r = 12%

$D_1 = 15 \times 1.08 = ₹16.20$

$P_0 = 16.20 / (0.12 - 0.08)$

Price = ₹405.00 per share

Suited to a mature market leader growing dividends steadily, in line with the economy.

EQUITY

Two-Stage Growth Model

$$P_0 = D_1[\,1 - ((1+g_1)/(1+r))^{n}\,] / (r - g_1) \;+\; P\square\,/(1+r)^{t}, \quad P\square = D_(n+1)/(r - g_2)$$

Value = PV of high-growth dividends + PV of the eventual steady-state selling price

1 Stage 1 — High Growth

Dividends grow fast at g_1 for a defined number of years (n).

2 Stage 2 — Stable Growth

Growth settles to a sustainable long-run rate g_2 forever after.

Common for Indian IT / consumer companies expected to out-grow the economy for a few years before maturing.

EQUITY

Two-Stage Growth Stock: Example

Given: D_0 = ₹20 · High growth g_1 = 15% for 3 yrs · Stable growth g_2 = 6% thereafter · Required return r = 13%

1

Forecast high-growth dividends

$D_1 = 20 \times 1.15 = ₹23.00$

$D_2 = 23 \times 1.15 = ₹26.45$

$D_3 = 26.45 \times 1.15 = ₹30.42$

2

PV of high-growth dividends @13%

$PV(D_1)=20.35$ $PV(D_2)=20.72$

$PV(D_3)=21.08$

Sum = ₹62.15

3

Terminal price at end of Yr 3

$D_4 = 30.42 \times 1.06 = ₹32.24$

$P_3 = 32.24/(0.13-0.06) =$ ₹460.61

4

PV of terminal price + total

$PV(P_3) = 460.61/1.13^3 =$ ₹319.22

$P_0 = 62.15 + 319.22$

P_0 = ₹62.15 + ₹319.22 = ₹381.37 per share

EQUITY

Two-Stage Growth Stock: Example

Given: D_0 = ₹8 · High growth g_1 = 20% for 2 yrs · Stable growth g_2 = 7% thereafter · Required return r = 15%

1

Forecast high-growth dividends

$D_1 = 8 \times 1.20 = ₹9.60$

$D_2 = 9.60 \times 1.20 = ₹11.52$

2

PV of high-growth dividends @15%

$PV(D_1) = 9.60/1.15 = ₹8.35$

$PV(D_2) = 11.52/1.15^2 = ₹8.71$

Sum = ₹17.06

3

Terminal price at end of Yr 2

$D_3 = 11.52 \times 1.07 = ₹12.33$

$P_2 = 12.33/(0.15-0.07) = ₹154.08$

4

PV of terminal price + total

$PV(P_2) = 154.08/1.15^2 = ₹116.51$

$P_0 = 17.06 + 116.51$

P_0 = ₹17.06 + ₹116.51 = ₹133.57 per share

RECAP

Summary — Bond & Stock Valuation Concepts

Concept	Formula	Terminologies
Coupon Bond	$P = \Sigma C/(1+r)^t + F/(1+r)^T$	Coupon rate, nominal value, redemption at par
Zero-Coupon Bond	$P = F/(1+r)^T$	Nominal (redemption) value; no coupon rate
Perpetual Bond	$P = C/r$	Coupon only — no redemption / maturity
Yield to Maturity	$YTM \approx [C+(F-P)/n] / [(F+P)/2]$	True yield vs. stated coupon rate
Zero Growth Stock	$P_0 = D_1/r$	Constant dividend, no growth
Constant Growth Stock	$P_0 = D_1/(r-g)$	Steady dividend growth, cost of equity
Two-Stage Growth Stock	$PV(\text{high-growth Div}) + PV(\text{terminal price})$	Growth-rate transition, terminal price

Bonds promise fixed, contractual cash flows (coupon + redemption). Stocks offer growth-linked, uncertain cash flows (dividends) with no redemption value.



Thank You

Bonds & Stocks — Meaning, Features & Terminologies

Valuation Module